

PROBLEM STATEMENT #37 | Retail, E-Commerce & Finance

Group Travel Expense Simplification Engine

1. Problem Context & Overview

A multi-currency expense sharing utility that records uneven group expenses, converts foreign currencies via daily exchange rates, and computes minimum-transaction debt settlement graphs.

Target Stakeholders / Actors: *Group Traveler, Trip Admin*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall compute the optimal debt settlement graph using min-flow algorithms to minimize total peer-to-peer payment transactions.

Sample Acceptance Criteria: Pass: Group balance settles to net zero with minimal payment edges; Fail: Sum of debts does not equal sum of credits.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: The settlement calculation engine must execute in under 100 ms for groups of up to 50 travelers with 500 expenses.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.